Customer Sales and Activation Performance Analysis with Monthly Revenue Prediction

Done By:

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Abstract

This project focuses on analyzing sales and service activation data within a telecommunications retail environment to extract meaningful business insights and support data-driven decision-making. The study aims to understand customer behavior, evaluate sales and activation performance, and assess customer value through both descriptive and predictive analytics. By integrating business intelligence tools with machine learning techniques, the project provides a comprehensive analytical approach to revenue and customer analysis.

The implementation involved extensive data cleansing and preparation using Power BI, including handling missing values, removing redundant attributes, and transforming variables for analysis. Descriptive statistics and interactive dashboards were developed to analyze sales performance, activation trends, payment behavior, and Monthly Recurring Charge (MRC) distribution. Separate dashboards were created for sales and activation data, followed by a merged dataset to study customers who both purchased products and activated services. Additionally, predictive modeling was conducted using RapidMiner to classify customers into High and Low MRC levels based on activation characteristics, sales behavior indicators, and market attributes, while carefully avoiding data leakage.

The results demonstrate that the dashboards successfully revealed key patterns in sales and activation behavior, highlighting high-value customer segments and revenue drivers. The predictive models achieved strong and realistic performance, with Logistic Regression and Random Forest producing high accuracy, precision, and recall in MRC classification. Clustering analysis further segmented customers into meaningful groups based on transaction value and behavior. Overall, the project confirms that combining business intelligence dashboards with predictive analytics provides valuable insights that can support strategic planning, targeted marketing, and revenue optimizat

Business Intelligence Project description and objectives

This project analyzes sales and service activation data in the telecommunications retail industry to support data-driven decision-making. Using Power BI, the project transforms raw transactional data into interactive dashboards that highlight sales performance, activation trends, and customer value. In addition, machine learning models built with RapidMiner are used to predict Monthly Recurring Charge (MRC) levels and segment customers based on their behavior.

The project helps telecommunications retailers better understand customer purchasing and activation patterns, identify high-value customers, and optimize revenue strategies. By combining business intelligence with predictive analytics, the study demonstrates how data-driven insights can improve business performance and strategic planning.

Goal:

To analyze sales and service activation data using business intelligence and machine learning techniques, and to develop a data-driven analytics framework that supports informed decision-making, customer value assessment, and recurring revenue optimization within a telecommunications retail environment.

Objectives:

Analyze Sales and Activation Performance: Examine sales transactions and service activation data to identify key trends, patterns, and performance indicators that impact revenue and customer behavior.

Integrate Sales and Activation Data: Merge sales and activation datasets to analyze customers who both activated services and purchased products, enabling cross-selling and bundled revenue analysis.

Develop Interactive BI Dashboards: Build dynamic dashboards using Power BI to visualize key metrics such as sales amount, activation volume, and Monthly Recurring Charge (MRC), supporting exploratory analysis and performance monitoring.

Understand Key Drivers of MRC: Identify and analyze the factors that influence customer MRC levels, including activation type, plan characteristics, product category, and market attributes.

Predict Customer MRC Level: Develop machine learning classification models using RapidMiner to predict whether customers belong to high or low MRC segments based on activation and sales behavior.

Enhance Decision-Making: Provide actionable, data-driven insights that support strategic decisions related to pricing, plan selection, cross-selling opportunities, and customer targeting.

Optimize Revenue Strategies: Utilize analytical findings and model results to recommend strategies aimed at increasing recurring revenue and improving customer value.

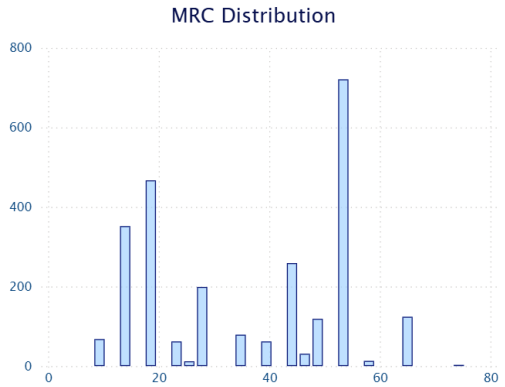
Data Description and understanding

marketid	custno	company	item	itmdesc	qty	cost	discount	Price	taxable	taxamount	adduser	name	lastname	invno	adddate	acttype	paytype
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	CSW71751	Lucero	Hidrogo	100666	Wednesday, October 1, 2025	Reactivation	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	WCM17599	America	Contreras	100691	Thursday, October 2, 2025	Upgrade	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	VBE07904	Alix	DimasReyes	100718	Saturday, October 4, 2025	Reactivation	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	WXA06719	Carlos	Landaverde	100771	Sunday, October 5, 2025	New Activation	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	CSW71751	Lucero	Hidrogo	100803	Tuesday, October 7, 2025	New Activation	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	VBE07904	Alix	DimasReyes	100888	Saturday, October 11, 2025	New Activation	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	WXA06719	Carlos	Landaverde	100922	Monday, October 13, 2025	Reactivation	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	WCM17599	America	Contreras	101001	Saturday, October 18, 2025	Upgrade	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	VBE07904	Alix	DimasReyes	101002	Saturday, October 18, 2025	Reactivation	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	VBE07904	Alix	DimasReyes	101012	Sunday, October 19, 2025	Upgrade	Cash
NC	TLNC-1406	NC SKYWAY	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	WCM17599	America	Contreras	101102	Thursday, October 23, 2025	Reactivation	Cash
NC	TLNC-2112	NC 2112 W	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	QDD75196	Johana	Ramos	93597	Saturday, October 4, 2025	Reactivation	Cash
NC	TLNC-2112	NC 2112 W	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	GRP93962	Arnoldo	Pulido	93625	Tuesday, October 7, 2025	Reactivation	Cash
NC	TLNC-2112	NC 2112 W	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	CXQ84041	Diana	Avaloss	93650	Thursday, October 9, 2025	Reactivation	Cash
NC	TLNC-2112	NC 2112 W	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	CXQ84041	Diana	Avaloss	93677	Sunday, October 12, 2025	Reactivation	Cash
NC	TLNC-2112	NC 2112 W	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	QDD75196	Johana	Ramos	93722	Saturday, October 18, 2025	Upgrade	Cash
NC	TLNC-2112	NC 2112 W	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	CXQ84041	Diana	Avaloss	93735	Sunday, October 19, 2025	Reactivation	Cash
NC	TLNC-2112	NC 2112 W	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	CXQ84041	Diana	Avaloss	93745	Monday, October 20, 2025	Reactivation	Cash
NC	TLNC-3101	NC FREEDOM	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	DUJ98602	Vilma	Guzman	120547	Wednesday, October 1, 2025	New Activation	Cash
NC	TLNC-3101	NC FREEDOM	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	JXU00539	Daniel	Lazcano	120549	Wednesday, October 1, 2025	New Activation	Cash
NC	TLNC-3101	NC FREEDOM	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	DUJ98602	Vilma	Guzman	120601	Friday, October 3, 2025	Reactivation	Cash
NC	TLNC-3101	NC FREEDOM	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	JXU00539	Daniel	Lazcano	120615	Friday, October 3, 2025	New Activation	Cash
NC	TLNC-3101	NC FREEDOM	LINEONLYACT	LINE ONLY ACTIVATION	1	0	0	0	F	0	DUJ98602	Vilma	Guzman	120637	Saturday, October 4, 2025	Reactivation	Cash

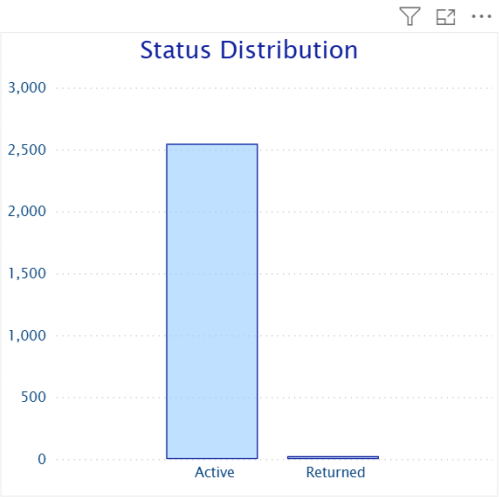
Field Name	Description	Importance to the Project
invno	Unique invoice number for each transaction	Used to link sales transactions with activation records during data merging.
marketid	Identifier of the market or store location	Helps analyze performance and behavior across different markets or branches.
custno	Unique customer identifier	Enables customer-level analysis and supports merging sales and activation data.
company	Name of the retail store	Useful for store-level performance comparison.
item	Code representing the sold item	Identifies specific products for sales analysis.
itmdesc	Description of the sold item	Helps classify products (phones, accessories, devices).
qty	Quantity of items sold	Used to analyze sales volume and demand.
cost	Cost of the item for the business	Important for understanding pricing structure and value analysis.
discount	Discount applied to the item	Help analyze promotional impact and pricing strategies.
realprice	Final selling price after discounts	Used to measure actual revenue from sales.
taxable	Indicates whether the item is taxable	Helps ensure accurate sales and tax analysis.
taxamount	Tax charged on the transaction	Important for understanding total transaction value.
adduser	Username of the employee who processed the sale	Allow employee-level performance analysis if needed.
name	Employee first name	Used for reference and internal reporting.

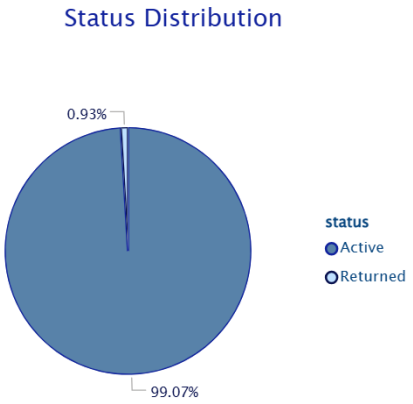
lastname	Employee last name	Complement employee identification.
adddate	Date and time of the sales transaction	Enables time-based analysis and trend evaluation.
paytype	Payment method used	Helps analyze customer payment behavior.
prodline	Product line category	Supports segmentation of products (e.g., ACCESSORY, PHONES).
category	High-level product category	Used to summarize and compare product performance.
subcategory	Detailed product classification	Enables deeper product-level insights.
cashpaid	Amount paid in cash	Used to calculate total sales amount and payment distribution.
creditcardpaid	Amount paid by credit card	Helps analyze preferred payment methods.
debitcardpaid	Amount paid by debit card	Complement payment behavior analysis.
tangible	Indicates if the item is a physical product	Useful for differentiating physical and non-physical sales.
serialized	Indicates if the item has a serial number	Help distinguish devices from non-serialized items.
manufacturer	Brand of the sold item	Enables brand-level sales analysis.
color	Color of the product	Useful for inventory and product preference analysis.
state	State or geographic location	Supports geographic performance analysis.
Sales Amount	Total sales value calculated from payments	Key financial metric for sales performance analysis.
actdate	Date and time of service activation	Used for activation trend and timing analysis.
status	Activation status (e.g., Active)	Help evaluate service lifecycle and activation success.
acttype	Type of activation (New, Upgrade, Reactivation)	Important for understanding customer lifecycle behavior.
grouptype	Account grouping (Individual or Group)	Helps segment customers by account structure.
plandesc	Description of the activated plan	Used to analyze plan popularity and offerings.
mrc	Monthly Recurring Charge value	Core variables are used to measure customer value and create MRC level (High/Low).

96K	37	40	10	75
Sum of mrc	Average of mrc	Median of mrc	Min of mrc	Max of mrc



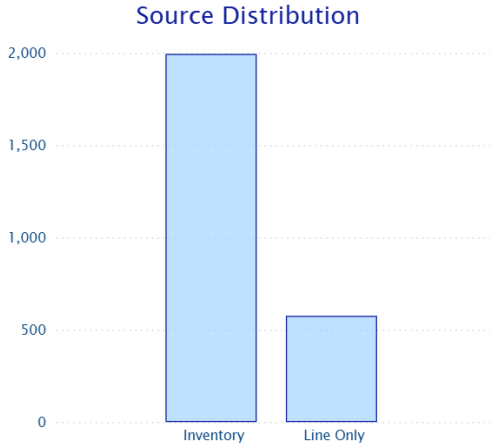
The MRC distribution is right-skewed, meaning most customers have lower monthly charges, while a smaller number pay higher MRC values. This is evident because the median (40) is slightly higher than the average (37) and the distribution have a long tail to the right, indicating fewer high-value plans that increase overall variability.

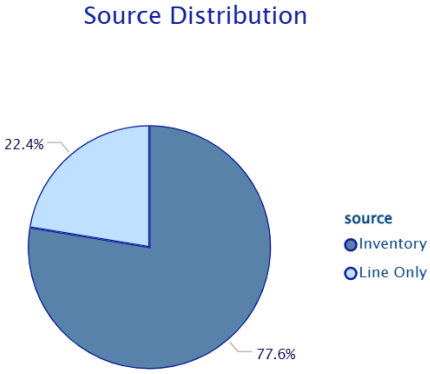




The status distribution shows that most records are Active (99.07%), while only a very small portion are Returned (0.93%). This indicates that most customers or lines remain active, with minimal returns.

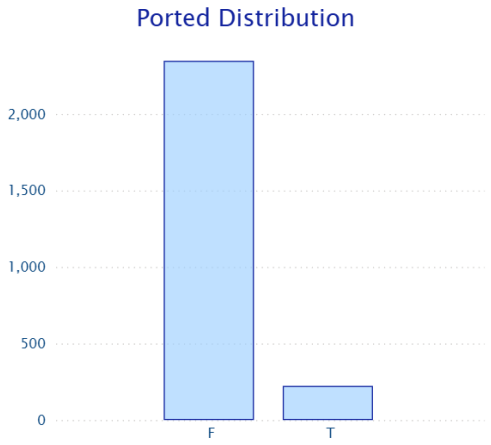
The strong imbalance suggests overall operational stability, but it also highlights that Returned cases are rare, which is important to consider in analysis and modeling, as minority classes may have limited impact unless handled carefully.

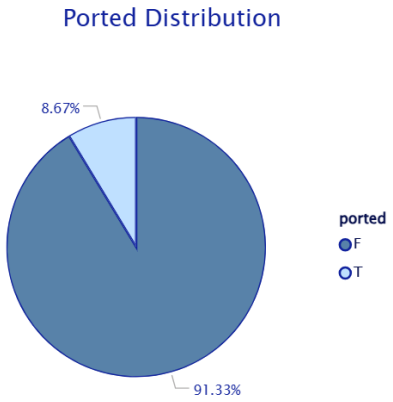




The source distribution shows that most transactions come from Inventory (77.6%), while Line Only transactions account for 22.4%. This indicates that most activations and sales are associated with physical inventory items, rather than line-only services.

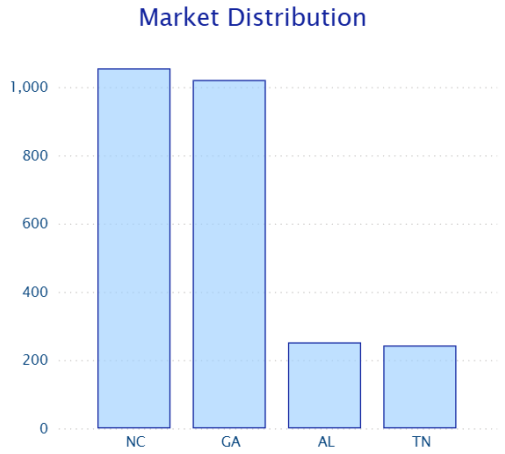
This pattern suggests that bundled device or accessory sales play a larger role in overall activity compared to standalone line activations.

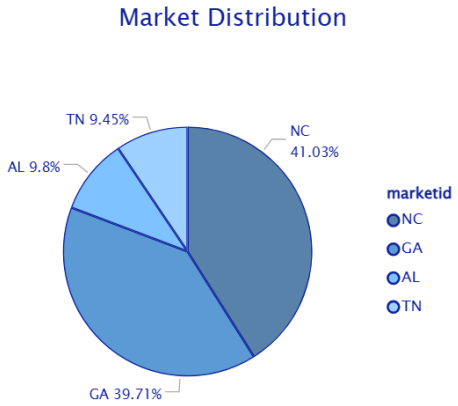




The ported distribution shows that most activations are not ported (F – 91.33%), while only 8.67% are ported (T) from another provider. This indicates that most customers are acquiring new numbers rather than switching from a different carrier.

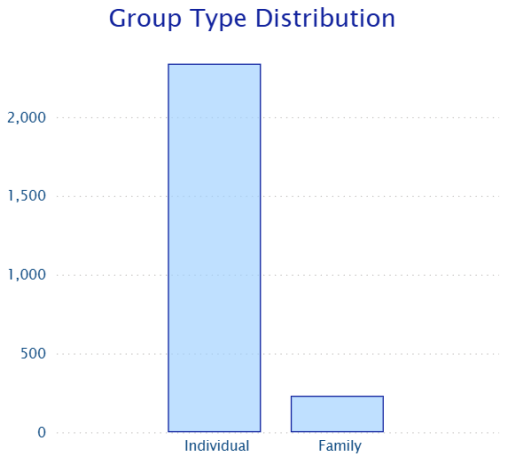
This suggests stronger performance in new customer acquisitions compared to competitor port-ins.

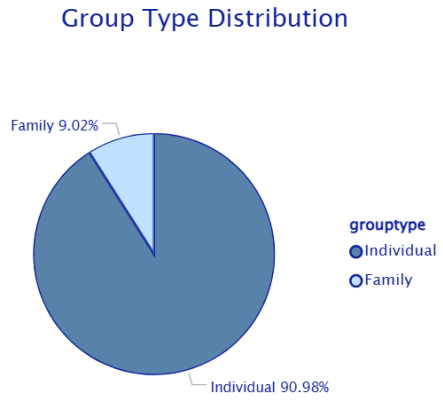




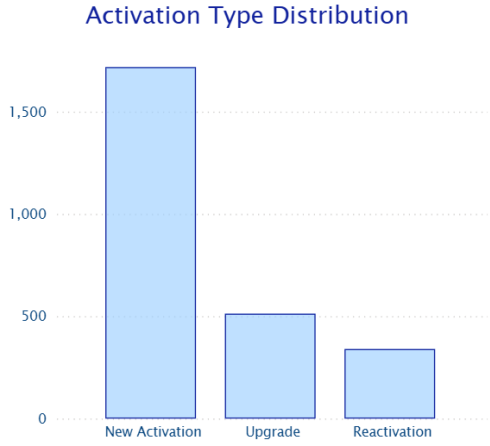
The market distribution shows that North Carolina (41.03%) and Georgia (39.71%) account for most activations, together representing over 80% of the total. In contrast, Alabama (9.8%) and Tennessee (9.45%) contribute a much smaller share.

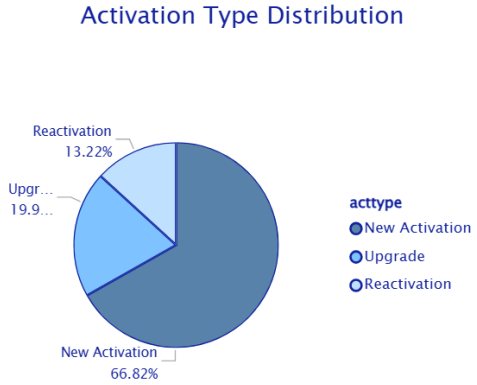
This indicates that business activity is highly concentrated in NC and GA, making them the key markets driving overall performance.





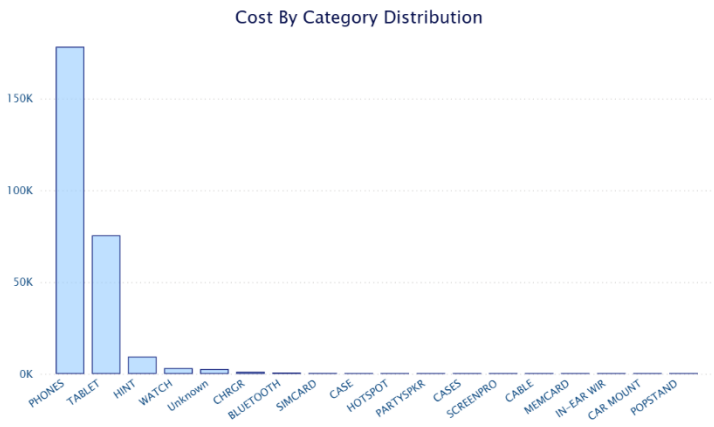
Most customers belong to the Individual group type (90.98%), while Family accounts represent only 9.02% of the total. This indicates that most activations and sales are driven by individual customers rather than family plans.





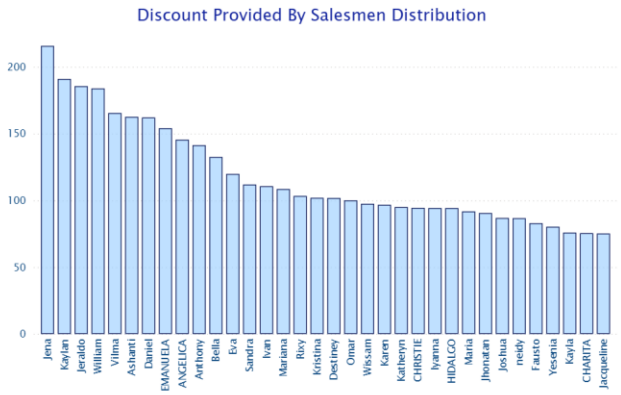
Most transactions are New Activations (66.82%), indicating strong customer acquisition. Upgrades account for 19.95%, reflecting existing customers enhancing their plans, while Reactivations represent 13.22%, showing a smaller portion of previously inactive customers returning.

272.59K	86.67	5.18	0.00	1.22K
Sum of cost	Average of cost	Median of cost	Min of cost	Max of cost



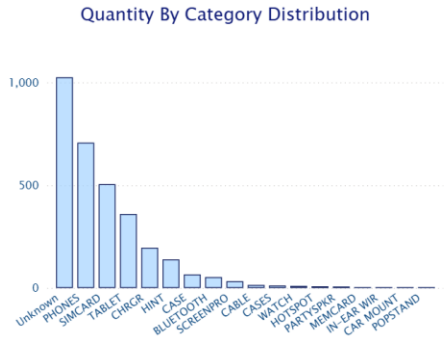
The cost analysis shows a total cost of 272.59K, with an average cost of 86.67 and a much lower median of 5.18, indicating that overall costs are heavily influenced by a small number of high-value items. The distribution highlights that Phones are the main cost driver, followed by Tablets, while the remaining categories contribute minimally. This suggests that the business's cost structure is concentrated around major device categories, making them critical for cost management and profitability optimization.

245.49K	78.06	10.00	0.00	580.00
Sum of discount	Average of discount	Median of discount	Min of discount	Max of discount



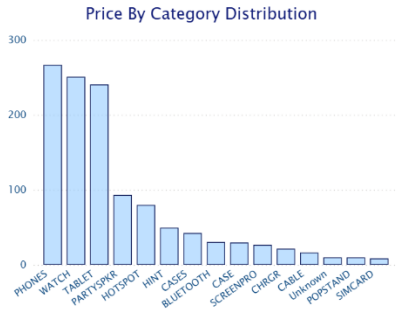
The discount analysis shows a total discount of 245.49K, with an average discount of 78.06 and a median of 10.00, indicating that most transactions receive relatively small discounts while a few large discounts raise the overall average. The distribution by salesperson highlights noticeable variation in discounting behavior, where a small number of salesmen account for higher total discounts compared to others. This suggests opportunities to standardize discount policies and monitor high-discount practices to protect margins while maintaining sales performance.

3K	0.98	1	-1	1
Sum of qty	Average of qty	Median of qty	Min of qty	Max of qty

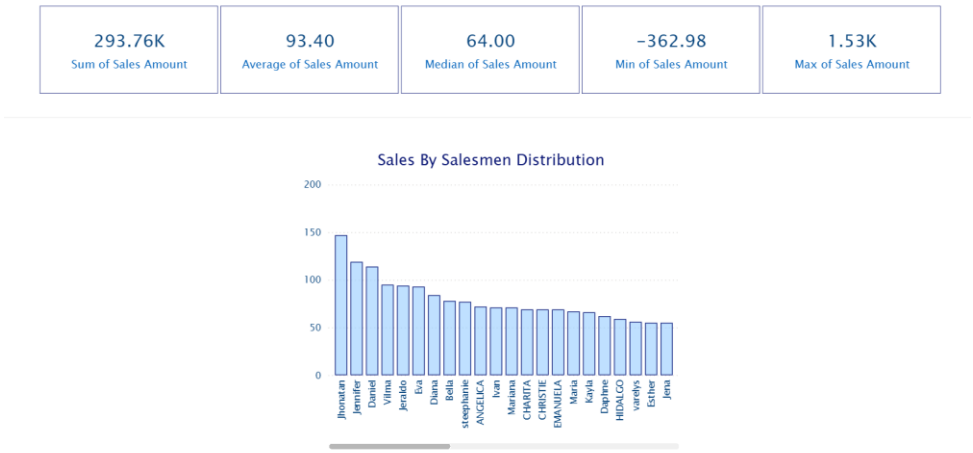


This quantity dashboard shows a total of about 3K items sold, with an average and median quantity close to 1, indicating that most transactions involve a single item. The category distribution reveals that Unknown, Phones, SIM Cards, and Tablets account for the highest sales volumes, while accessories such as cables, mounts, and stands contribute relatively small quantities. Overall, the results highlight a sales pattern dominated by core telecom products with low per-transaction quantities, supporting the focus on high-volume essential items.

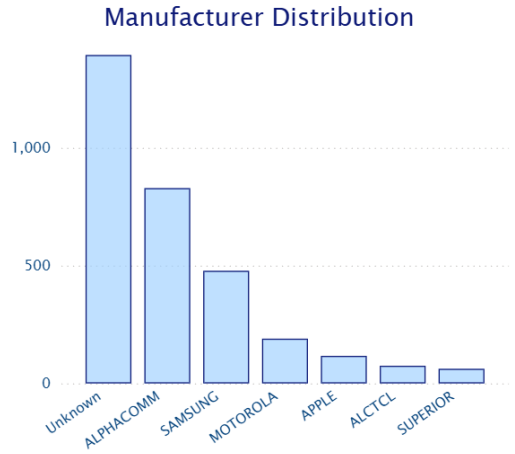
313.77K	99.77	29.99	0.00	1.20K
Sum of Price	Average of Price	Median of Price	Min of Price	Max of Price

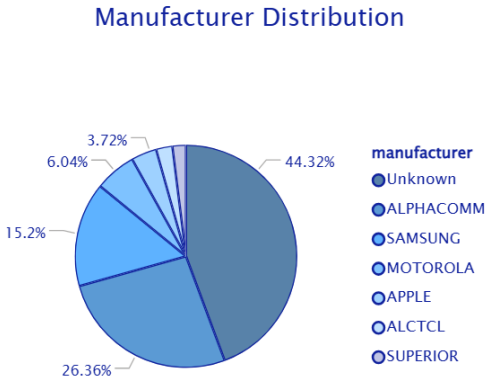


This dashboard summarizes the price distribution by product category, showing a total sales value of 313.77K, with an average price of 99.77 and a median of 29.99, indicating that most transactions are low-priced items while a few high-value products raise the average. Phones, watches, and tablets contribute the highest total prices, reflecting their strong revenue impact, whereas accessories such as cases, cables, SIM cards, and popstands have much lower price contributions. The maximum price reaches 1.20K, highlighting the presence of premium devices, while the minimum price of 0.00 suggests promotional or bundled items. Overall, the distribution is right-skewed, where high-end devices drive revenue and low-cost accessories support volume sales.

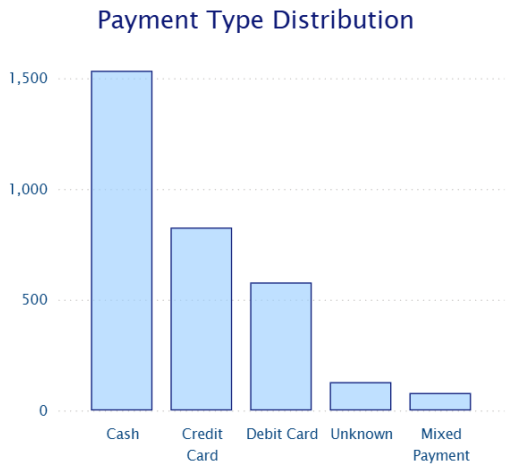


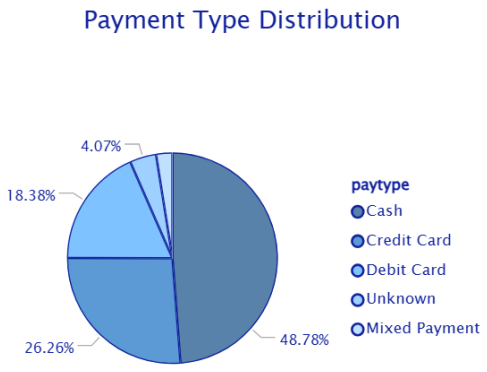
This dashboard summarizes overall sales performance and salesperson contribution. Total sales reached 293.76K, with an average transaction value of 93.40 and a median of 64.00, indicating that most sales are moderate in value while a few high-value transactions increase the average. The negative minimum sales amount (-362.98) suggests refunds or returns, while the maximum (1.53K) highlights occasional high-ticket sales. The bar chart shows sales distribution by salesperson, where a small group of top performers generates higher sales, followed by a gradual decline across others, indicating performance variation and opportunities for targeted coaching or incentive strategies.



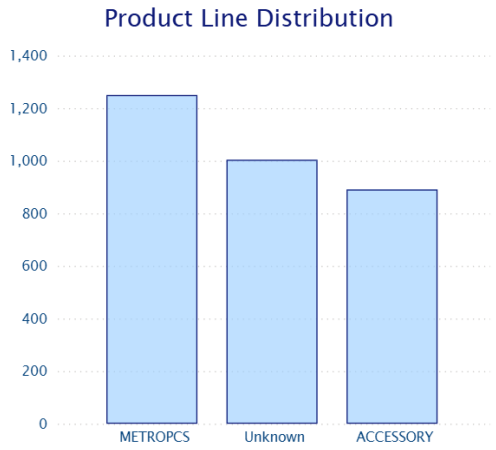


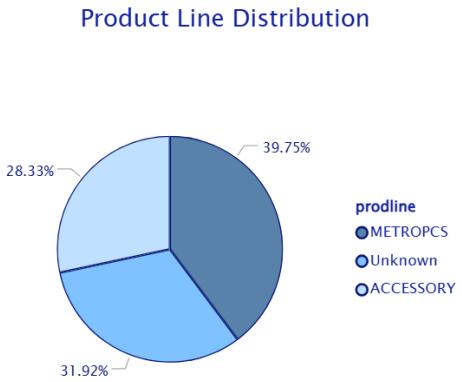
The manufacturer data is highly concentrated. Unknown (~44.3%) dominates the dataset, followed by ALPHACOMM (~26.4%) and SAMSUNG (~15.2%). Other manufacturers (Motorola, Apple, ALCTCL, Superior) each contribute relatively small shares, indicating limited diversity outside the top manufacturers.



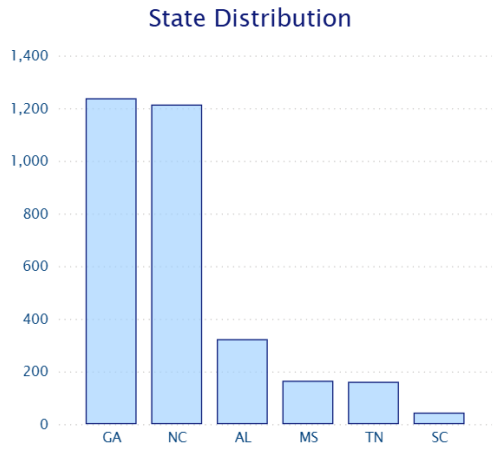


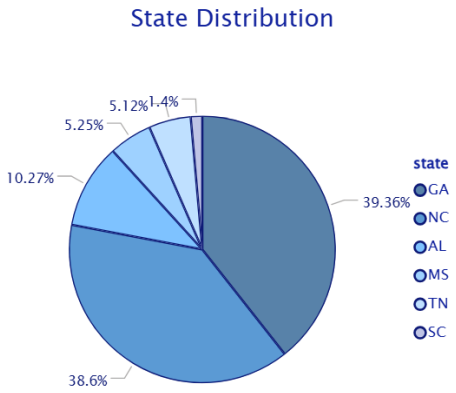
Payments are dominated by Cash (~48.8%), followed by Credit Card (~26.3%) and Debit Card (~18.4%). Unknown and Mixed Payment methods account for a small share, indicating that most transactions rely on traditional single-payment methods, with cash being the preferred option.





Sales are mainly driven by METROPCS products (~39.8%), making it the largest product line. Unknown (~31.9%) also represents a significant portion, indicating incomplete or unspecified product line information in many transactions. Accessories (~28.3%) contribute a substantial share as well, showing strong demand for add-on products alongside core offerings.



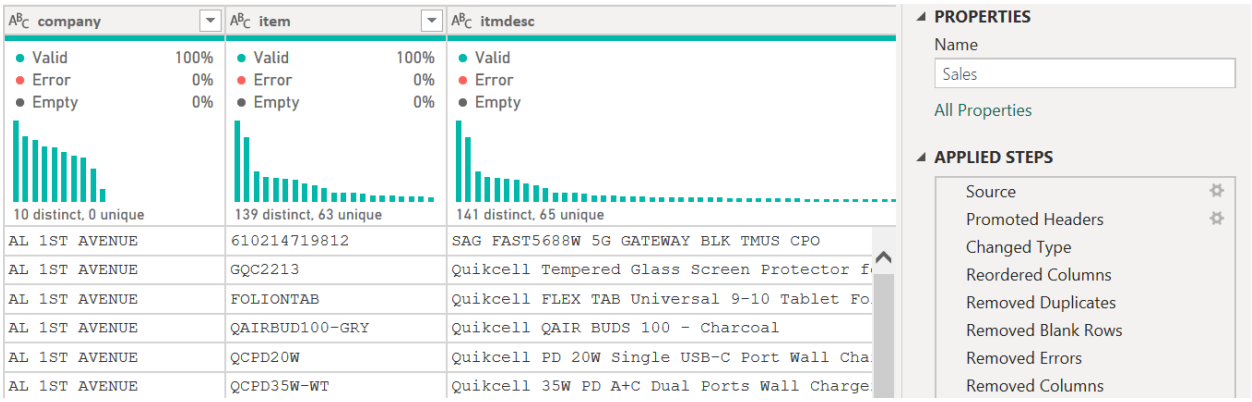


Sales are highly concentrated in Georgia (GA ~39.4%) and North Carolina (NC ~38.6%), which together account for most transactions. Alabama (AL ~10.3%) contributes a moderate share, while Mississippi (MS), Tennessee (TN), and South Carolina (SC) each represent relatively small portions of total sales.

Perform initial Data Exploratory analysis EDA with charts/graphs and record your understanding of what insights and patterns you have found on data, and why it relates to your project]

Data Primary Cleaning and Transformation

is the process of identifying and correcting errors, inconsistencies, or missing values in a dataset to improve its quality and reliability. It includes removing duplicates, fixing incorrect entries, handling missing data, and formatting values properly. This step ensures that the data is accurate, consistent, and ready for meaningful analysis and modeling.



Data cleansing was performed using Power BI (Power Query) to ensure data accuracy, consistency, and readiness for analysis. The following steps were applied:

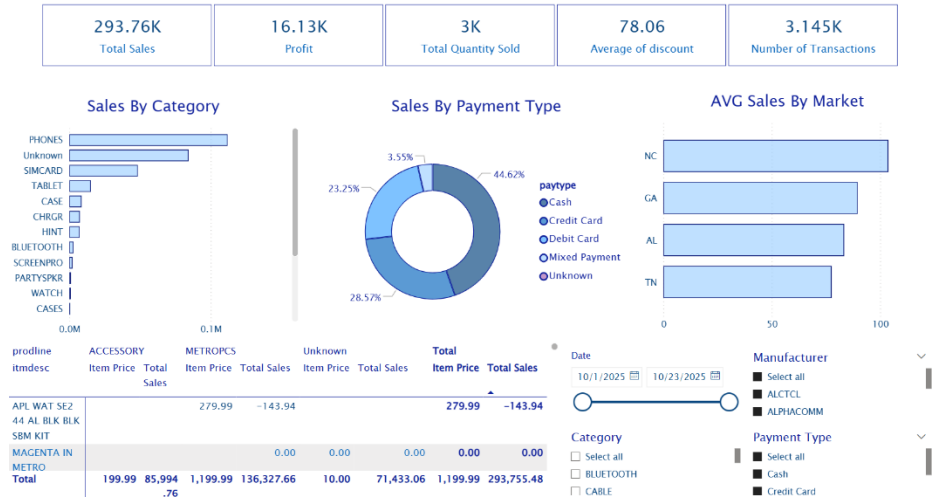
- Promoted Headers to ensure that the first row of the dataset was correctly recognized as column names.
- Changed Data Types to assign appropriate data types to each column (text, numeric, and date) for accurate calculations and analysis.
- Reordered Columns to improve dataset structure and readability.
- Removed Duplicate Records to prevent repeated transactions from affecting analytical results.
- Removed Blank Rows to eliminate incomplete records with missing values.
- Removed Errors to ensure that all remaining records were valid and analyzable.
- Removed Unnecessary Columns that were not relevant to the analysis, such as redundant identifiers and non-informative attributes.

As a result of these steps, all retained columns contained 100% valid values with no errors or empty records, ensuring a clean and reliable dataset for descriptive analytics, dashboard development, clustering, and predictive modeling.

Data Visualization and Insights

Power BI Dashboards:

Sales Dashboard



Key Performance Indicators (Top Cards)

- Total Sales (293.76K): Total revenue generated from all transactions.
- Profit (16.13K): Net profit after costs and discounts.
- Total Quantity Sold (3K): Number of items sold.
- Average Discount (78.06): Average discount applied across sales.
- Number of Transactions (3.145K): Total completed sales transactions.

Sales by Category

- Displays total sales by product category.
- Phones are the top-selling category.
- Accessories like SIM cards, tablets, cases, and chargers contribute to smaller portions.

Shows which products drive the most revenue.

Sales by Payment Type

- Donut chart showing sales distribution by payment method.
- Cash and Credit Card dominate transactions.
- Debit Card and Mixed Payments have moderate usage.
- Unknown payment types are minimal.

Help understand customer payment behavior.

Average Sales by Market

- Compares average sales across markets.
- NC and GA lead in average sales.
- AL and TN show lower performance.

Useful for identifying strong and weak regional markets.

Detailed Sales Table

- Shows item-level data including:
    - Product line and description
    - Item price
    - Total sales by category (Accessory, MetroPCS, Unknown)
- Helps trace totals back to individual products.

Supports detailed operational analysis.

Filters and Interactivity

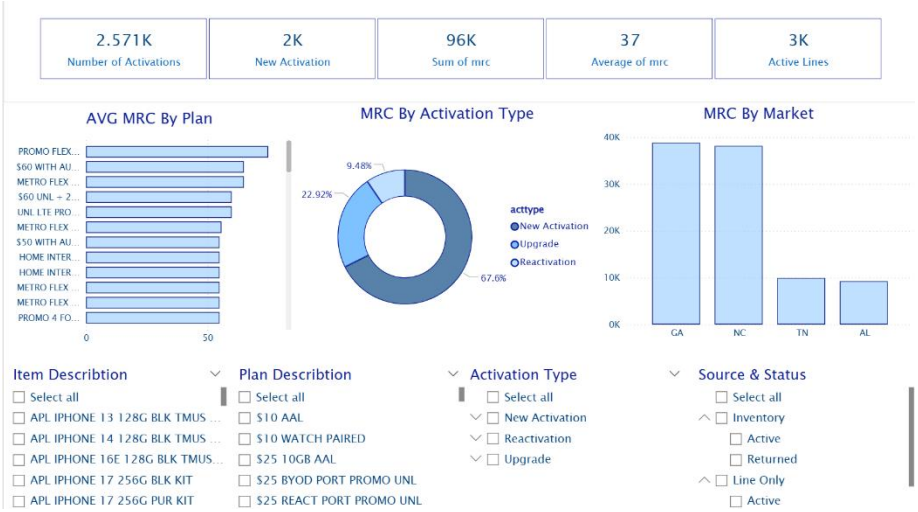
- Date range, Category, Manufacturer, and Payment Type slicers.
- Allows users to drill down and analyze specific periods, products, or markets.

Overall Insight

This dashboard provides a complete view of sales performance, showing:

- Phones as the main revenue driver
- Cash and card as preferred payment methods
- Stronger performance in NC and GA markets
- Healthy transaction volume but relatively moderate profit

Activation Dashboard



Key KPIs (Top Cards)

- Number of Activations (2.571K): Total activations processed in the selected period.
- New Activations (2K): Majority of activations are new customer lines.
- Sum of MRC (96K): Total monthly recurring charges generated.
- Average MRC (37): Average monthly charge per active line.
- Active Lines (3K): Total currently active service lines.

These KPIs summarize customer growth and recurring revenue performance.

Average MRC by Plan

- Shows average monthly charge by plan type.
- Promo Flex and Unlimited plans have the highest average MRC.
- Lower-tier and promotional plans generate less MRC.

Helps identify which plans contribute most to recurring revenue.

MRC by Activation Type

- Donut chart breaks MRC by:
    - New Activations (≈67.6%) – largest share
    - Upgrades (≈22.9%)
    - Reactivations (≈9.5%)

Indicating revenue growth is mainly driven by new customers.

MRC by Market

- Compare recurring revenue across markets.
- GA and NC generate the highest MRC.
- TN and AL contribute significantly less.

Useful for regional performance and target growth markets.

Filters & Interactivity

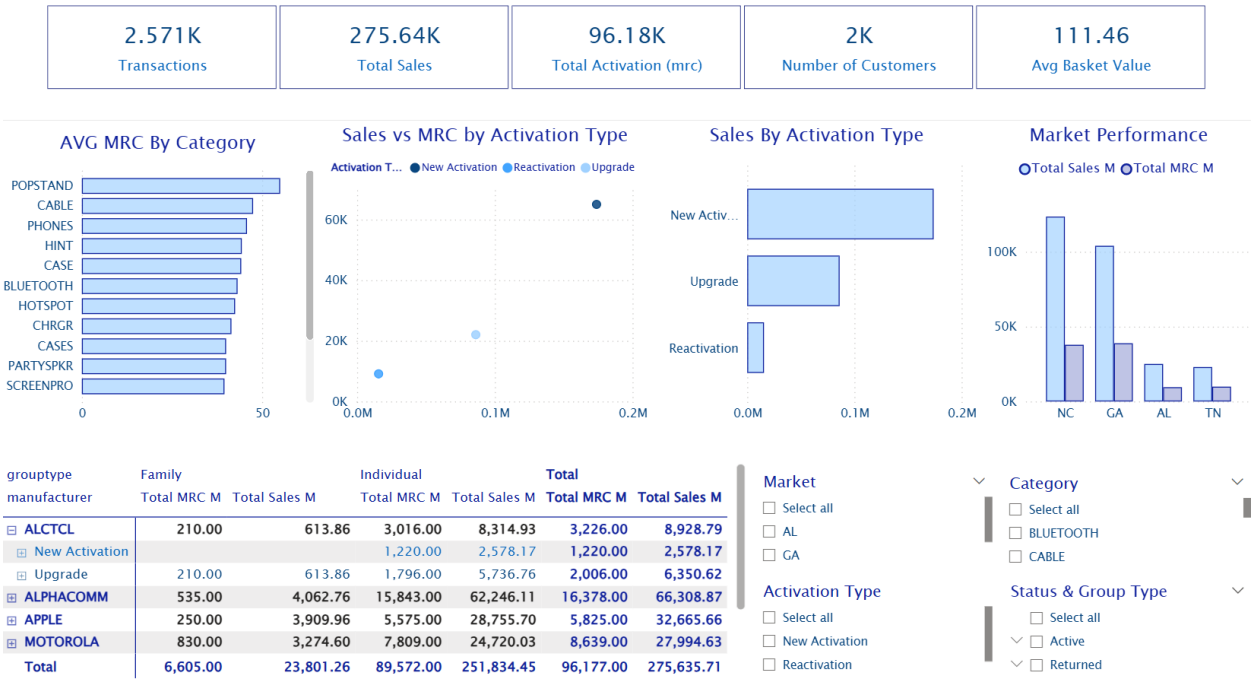
- Slicers for Item Description, Plan Description, Activation Type, and Source & Status.
- Enable deep analysis by plan, product, customer status, or activation source.

Overall Insight

This dashboard focuses on recurring revenue and customer activations, showing that:

- New activations are the main revenue driver
- Premium and promo plans yield higher MRC
- GA and NC are the strongest markets
- Active line base is healthy and growing

Sales & Activation Dashboard



grouptype	Family		Individual		Total	
manufacturer	Total MRC M	Total Sales M	Total MRC M	Total Sales M	Total MRC M	Total Sales M
ALCTCL	210.00	613.86	3,016.00	8,314.93	3,226.00	8,928.79
New Activation			1,220.00	2,578.17	1,220.00	2,578.17
Upgrade	210.00	613.86	1,796.00	5,736.76	2,006.00	6,350.62
ALPHACOMM	535.00	4,062.76	15,843.00	62,246.11	16,378.00	66,308.87
APPLE	250.00	3,909.96	5,575.00	28,755.70	5,825.00	32,665.66
MOTOROLA	830.00	3,274.60	7,809.00	24,720.03	8,639.00	27,994.63
Total	6,605.00	23,801.26	89,572.00	251,834.45	96,177.00	275,635.71

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Top KPI Cards (Overall Performance)

- Transactions (2.571K): Total number of completed sales transactions.
- Total Sales (275.64K): Overall revenue generated.
- Total Activation (MRC) – 96.18K: Total monthly recurring charges from activations.
- Number of Customers (2K): Unique customers served.
- Avg Basket Value (111.46): Average spend per transaction.

These KPIs give a quick snapshot of sales volume, revenue strength, and customer value.

AVG MRC by Category

- Shows average monthly recurring charge by product category.
- Categories like POPSTAND, CABLE, and PHONES have higher average MRC.
- Lower categories (e.g., SCREENPRO, PARTYSPKR) contribute less recurring value.

Help identify which product types generate stronger recurring revenue.

Sales vs MRC by Activation Type (Scatter Plot)

- Compare Total Sales (X-axis) vs Total MRC (Y-axis).
- New Activations dominate both sales and MRC.
- Upgrades contribute moderate sales and MRC.
- Reactivations have the smallest impact.

It shows that new activations are the main growth drive.

Sales by Activation Type

Horizontal bars summarize sales contribution:

- New Activation = highest sales
- Upgrade = mid-level
- Reactivation = minimal

Confirm sales depend mostly on acquiring new customers.

Market Performance

- Compares Total Sales vs Total MRC across markets.
- NC and GA are the strongest markets.
- AL and TN contribute much less.

Useful for regional performance and market prioritization.

Detailed Manufacturer Table

Breaks down MRC and Sales by:

- Manufacturers (Apple, Motorola, Alphacomm, etc.)
- Group type (Family vs Individual)
- Activation type

Apple and Alphacomm generate the highest combined sales and MRC.

Supports product and vendor performance analysis.

Filters (Interactivity)

Market, Category, Activation Type, and Status filters allow focused analysis.

Enables users to drill down by region, product, or activation behavior.

Overall Insight

This dashboard connects sales performance with recurring revenue, showing that:

- New activations drive most revenue,
- Phones and core accessories have higher MRC,
- NC and GA markets perform best,
- Apple and Alphacomm are top contributors.

Advanced Analytics and AI Modeling

To predict Monthly Recurring Charge (MRC) levels, the following supervised learning models were implemented using RapidMiner:

Logistic Regression:

Chosen for its interpretability and effectiveness in classification tasks. This model allows clear analysis of how key variables such as activation type, plan description, market, and sales amount influence the probability of a customer belonging to a specific MRC level.

Random Forest:

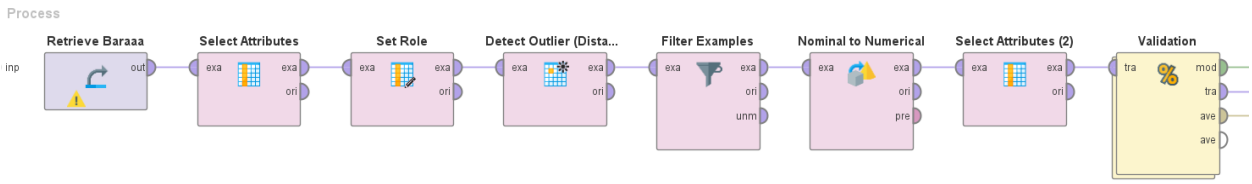
Selected due to its robustness, high predictive accuracy, and ability to handle complex relationships between variables. By combining multiple decision trees, Random Forest reduces overfitting and improves generalization when predicting MRC levels.

Unsupervised Model

Clustering analysis was conducted using the K-Means algorithm to uncover distinct employee segments. The Elbow Method, implemented in Python, was used to determine the optimal number of clusters (k=4), ensuring meaningful segmentation based on Job Satisfaction, Training Times Last Year, Performance Rating, and Attrition status.

Predictive Analysis

Building The Model:



Retrieve Data

Loads the dataset into RapidMiner from the repository. This is the starting point of the analysis.

Select Attributes

Keeps only the relevant variables (such as sales, plan, activation type, market, and product category) and removes unnecessary columns to reduce noise.

Set Role

Defines the target variable (label) as *MRC Level (High/Low)* and assigns any ID attributes. This step tells RapidMiner what the model should predict.

Detect Outlier (Distance-based)

Identifies unusual or extreme records that could negatively affect model performance. This improves prediction accuracy and model stability.

Filter Examples

Removes unwanted records (such as missing, invalid, or irrelevant values) to ensure clean and consistent data.

Nominal to Numerical

Converts categorical variables (e.g., activation type, plan description, market) into numerical form so they can be used by machine learning algorithms. (not included in Random forest because it can handle non-numerical columns)

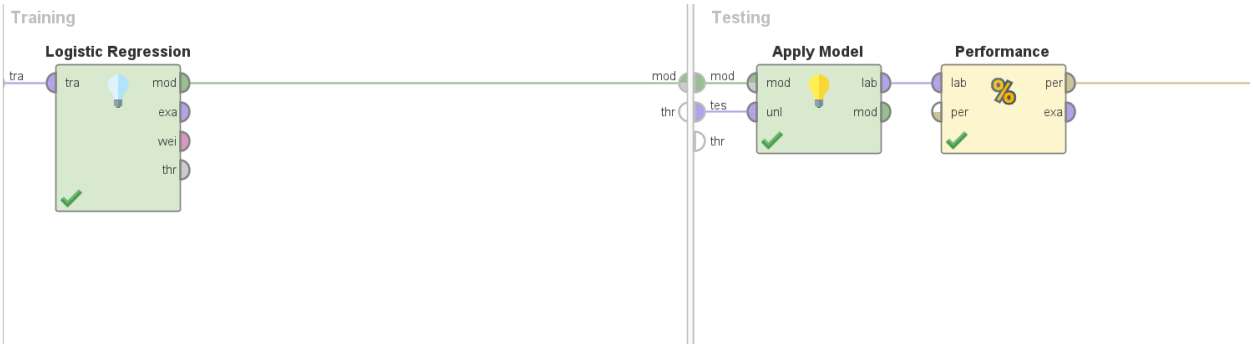
Select Attributes (Final)

Performs a final feature selection to keep the most important predictors and avoid overfitting.

Validation

Trains and tests the model (e.g., Random Forest) using validation techniques to evaluate performance. Outputs include model accuracy and prediction results.

Logistic Regression:



Training Phase (0.7): The Logistic Regression operator trains a model using labeled training data to learn patterns related to employee attrition.

Testing Phase (0.3): The Apply Model operator applies the trained model to unseen (unlabeled) test data to make predictions.

Evaluation Phase: The Performance operator compares the predicted labels with actual labels to assess the model's accuracy, precision, recall, and other performance metrics.

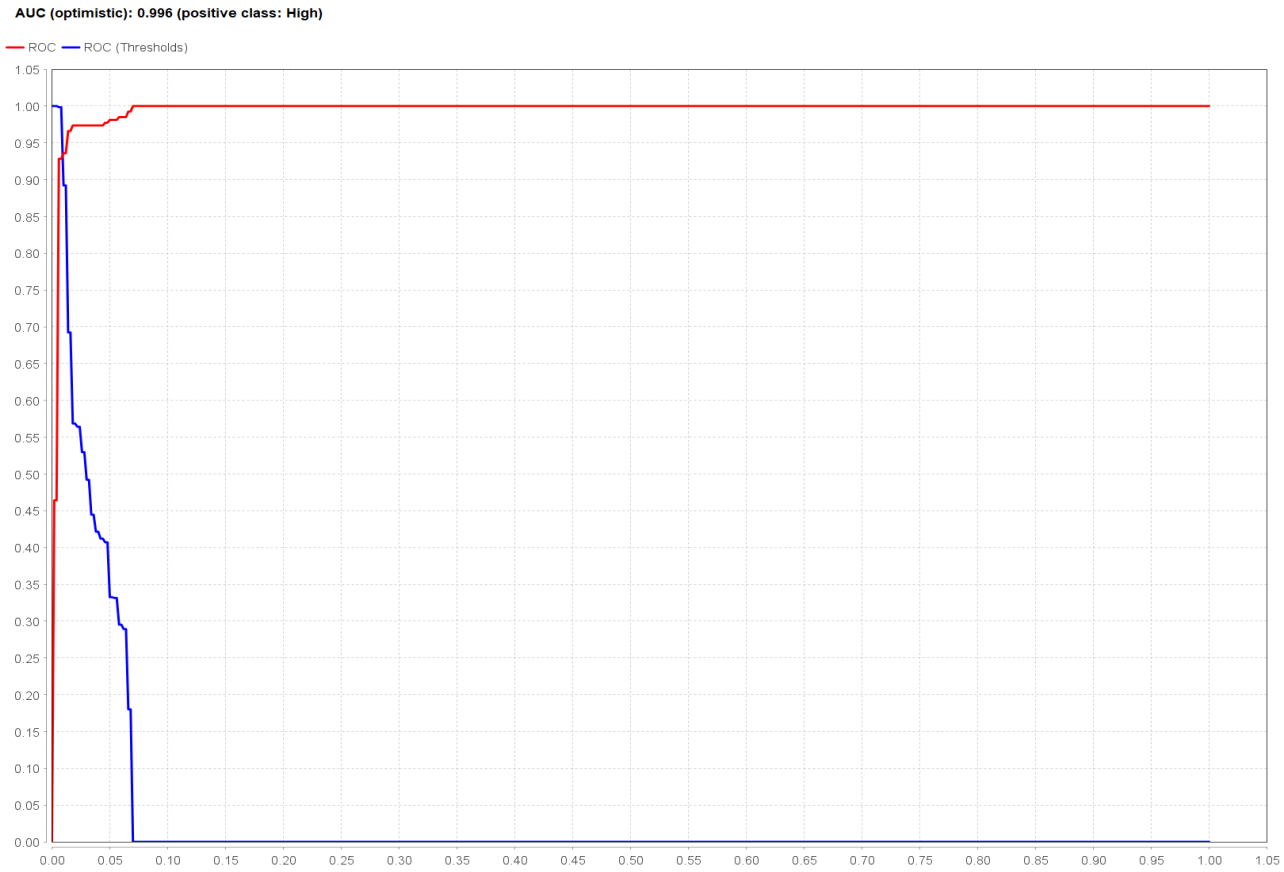
accuracy: 97.66%

	true Low	true High	class precision
pred. Low	242	7	97.19%
pred. High	5	258	98.10%
class recall	97.98%	97.36%	

The Random Forest model achieved an accuracy of 96.68%, demonstrating strong performance in classifying MRC levels as High or Low.

The confusion matrix shows that most records were correctly classified, with only a small number of misclassifications.

Both precision and recall are above 96% for the High and Low classes, indicating that the model predicts MRC levels reliably and maintains a good balance between correctly identifying High- and Low-MRC cases.

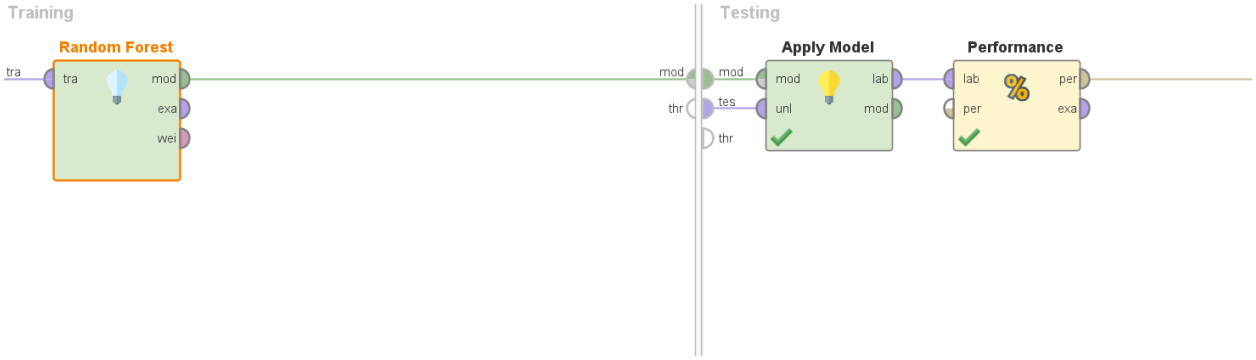


The ROC curve shows the trade-off between the true positive rate and false positive rate for different classification thresholds when predicting High vs Low MRC levels.

The model achieved an AUC value of 0.996, which indicates excellent discriminatory power. This means the Logistic Regression model is highly effective at distinguishing between customers with High MRC and Low MRC.

An AUC close to 1.0 suggests that the model performs very well across all thresholds, with minimal misclassification, making it reliable for predicting MRC levels.

Random Forest:



Training Phase (0.7): The Random Forest operator trains a model using the labeled training data to recognize complex patterns associated with employee attrition.

Testing Phase (0.3): The Apply Model operator uses the trained Random Forest model to predict attrition on unseen test data.

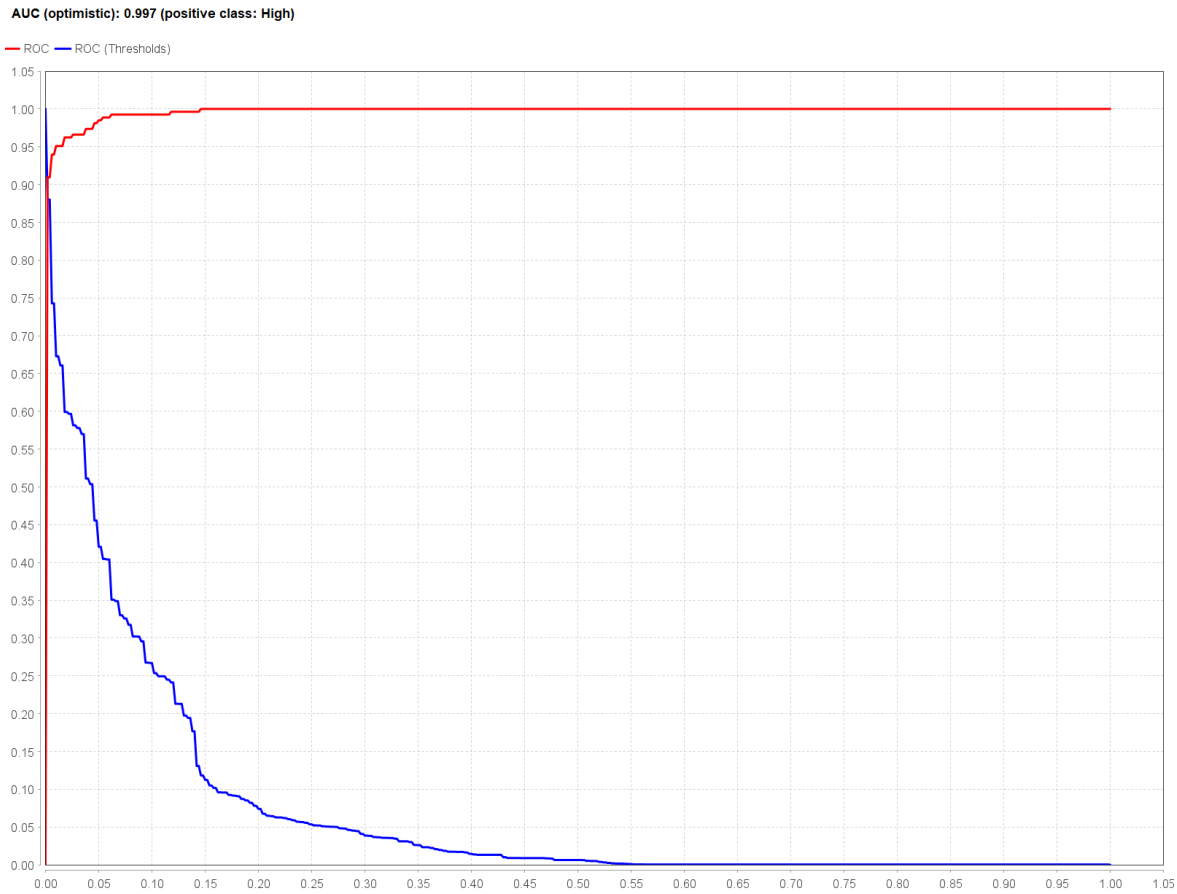
Evaluation Phase: The Performance operator evaluates the model by comparing predicted and actual labels, providing metrics such as accuracy, precision, and recall.

accuracy: 96.68%

	true Low	true High	class precision
pred. Low	237	7	97.13%
pred. High	10	258	96.27%
class recall	95.95%	97.36%	

The Random Forest model achieved an accuracy of 96.68%, indicating strong overall performance in classifying High and Low MRC levels.

- For Low MRC, the model correctly identified 237 cases, with a recall of 95.95% and precision of 97.13%, showing reliable detection of low-value customers.
- For High MRC, it correctly classified 258 cases, achieving a recall of 97.36% and precision of 96.27%, meaning most high-value customers were accurately identified.



The ROC–AUC value of 0.997 indicates an excellent ability of the Random Forest model to distinguish between High and Low MRC levels.

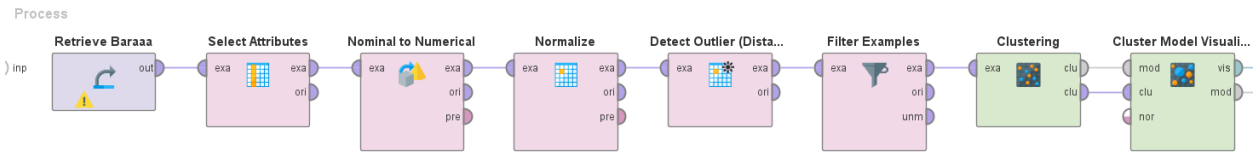
The ROC curve is very close to the top-left corner, which means the model achieves a high true positive rate with a very low false positive rate.

This confirms that the Random Forest model separates High and Low MRC cases extremely well across different decision thresholds, showing strong overall classification performance.

Clustering:

is a machine learning method that groups similar data points into clusters without using labels.

It was used to segment employees based on satisfaction, training, and performance — helping HR create targeted retention strategies for each group.



Retrieve Data
The dataset is loaded into RapidMiner as the starting point for analysis.

Select Attributes
Only relevant variables are chosen for clustering, removing unnecessary or irrelevant columns to improve cluster quality.

Nominal to Numerical
Categorical (nominal) attributes are converted into numerical form so they can be used by the clustering algorithm.

Normalize
All numerical attributes are scaled to the same range to prevent variables with large values from dominating the clustering results.

Detect Outliers (Distance-Based)
Extreme or abnormal data points are identified to reduce their impact on cluster formation.

Filter Examples
Detected outliers are removed, ensuring cleaner and more reliable clusters.

Clustering (K-Means)
The cleaned and normalized data is grouped into clusters based on similarity, revealing natural patterns in the data.

Cluster Model Visualization
The final step visualizes the clusters, helping interpret and understand the characteristics of each group.

Number of Clusters: 3

Cluster 0 1,438

prodline = Unknown is on average **100.00%** smaller, **acttype = Reactivation** is on average **93.17%** smaller, **prodline = ACCESSORY** is on average **73.20%** larger

Cluster 1 986

prodline = Unknown is on average **159.63%** larger, **acttype = Reactivation** is on average **149.68%** larger, **paytype = Unknown** is on average **104.46%** larger

Cluster 2 136

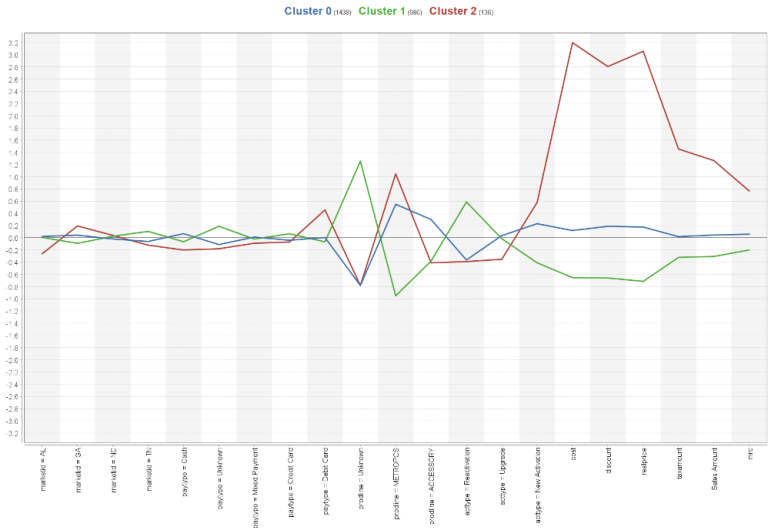
cost is on average **500.07%** larger, **realprice** is on average **421.34%** larger, **discount** is on average **406.35%** larger

Cluster	marketi...	marketi...	marketi...	marketi...	paytyp...	paytyp...	paytyp...	paytyp...	paytyp...	prodlin...	prodlin...	prodlin...	acttype...	acttype...	acttype...
Cluster 0	0.021	0.043	-0.020	-0.060	0.068	-0.111	0.012	-0.041	0.004	-0.782	0.549	0.300	-0.363	0.035	0.231
Cluster 1	0.005	-0.091	0.025	0.105	-0.065	0.189	-0.021	0.066	-0.067	1.256	-0.951	-0.382	0.587	-0.015	-0.410
Cluster 2	-0.261	0.194	0.039	-0.121	-0.200	-0.179	-0.089	-0.069	0.457	-0.782	1.048	-0.411	-0.390	-0.352	0.579

The analysis grouped the data into three distinct clusters based on product line, activation type, payment type, and pricing behavior etc..

- Cluster 0 (largest group) represents regular transactions, mainly accessory products with fewer reactivations. This group shows stable and typical customer behavior.
- Cluster 1 (medium group) is characterized by reactivations and unknown product or payment types, indicating less predictable behavior and possible data inconsistencies.
- Cluster 2 (smallest group) contains high-value transactions, with significantly higher cost, real price, and discounts. Although small, this cluster contributes the most in terms of revenue impact.

Overall, clustering helps distinguish regular customers, uncertain/reactivated cases, and high-value customers, supporting better targeting and decision-making.



These visualizations summarize the characteristics of the three clusters identified by the model.

- Cluster 0 (largest group) shows average behavior across most variables. It is mainly associated with the ACCESSORY product line and fewer Reactivation activities, indicating typical, low-risk customers.
- Cluster 1 (medium size) is characterized by higher proportions of Unknown product line, Reactivation account type, and Unknown payment type, suggesting less stable or incomplete customer profiles.
- Cluster 2 (smallest group) clearly stands out with much higher cost, real price, and discount values, representing high-value or high-cost transactions.

Overall, the heatmap and line chart highlight how Cluster 2 is financially intensive, while Clusters 0 and 1 differ mainly in categorical behaviors, helping distinguish customer segments effectively

Cluster 0 – Standard Accessory Customers

Insights:

- Dominated by the ACCESSORY product line.
- Lower presence of Reactivation and Unknown categories.
- Financial variables (cost, discount, real price) are close to the average.

Interpretation:
This cluster represents regular, stable customers with typical purchasing behavior and moderate transaction values.

Cluster 1 – Reactivation & Uncertain Profile Customers

Insights:

- Higher occurrence of Reactivation account type.
- More Unknown product lines and payment types.
- Moderate financial values without extreme costs or prices.

Interpretation:

This group reflects customers with incomplete or less consistent profiles, often returning or reactivated, and may require targeted engagement strategies.

Cluster 2 – High-Value Transactions Customers

Insights:

- Significantly higher cost, real price, and discount values.
- Financial behavior clearly separates this cluster from the others.
- Less influenced by categorical variables compared to monetary ones.

Interpretation:
This cluster represents high-value or premium transactions, potentially high-revenue customers but also higher risk due to large discounts and costs.

Tools Research and Selection Effort

Several tools were reviewed to support this project, focusing on data preparation, analysis, modeling, and visualization. After evaluating different options, **Power BI** was chosen for data cleaning, descriptive analytics, and interactive dashboard development because it allows efficient data transformation, dynamic visual exploration, and clear presentation of sales and activation insights. For predictive analytics, **RapidMiner** was selected due to its user-friendly visual workflow, strong support for classification algorithms, and built-in evaluation metrics, which made it suitable for building and comparing Logistic Regression and Random Forest models without extensive coding. Additionally, **Excel** was used for initial data inspection and validation. Together, these tools complemented each other by enabling end-to-end analysis—from raw data exploration to predictive modeling and actionable business insights—within a telecommunications retail context.

Project Deployment Effort – Use Case

In a real business environment, this project can be used as a decision-support system for sales and operations managers within a telecommunications retail company. The interactive Power BI dashboards allow managers to monitor sales performance, activation trends, product categories, and employee contributions in real time. By reviewing metrics such as total sales amount, average transaction value, and salesperson performance, management can quickly identify top performers, underperforming stores, and high-value products, enabling timely corrective actions.

The predictive model that classifies customers into High and Low MRC levels helps the business improve revenue optimization and targeted marketing. Sales teams can focus upselling efforts on customers with high MRC potential, while retention strategies can be applied to low MRC customers to increase their long-term value. This reduces random promotions and replaces them with data-driven customer targeting.

Additionally, combining sales and activation data helps the business understand cross-selling behavior, such as which products are most often purchased during service activation. This insight supports better product bundling, inventory planning, and staff training. Overall, the project leads to higher revenue efficiency, improved sales productivity, better customer segmentation, and more informed strategic planning, giving the business a competitive advantage in a highly competitive telecom market..

Results

The results of this project reveal clear patterns in sales performance, service activations, and customer value within the telecommunications retail environment. Through Power BI dashboards, it was observed that total sales reached approximately 293K, with an average transaction value of 93.4 and a median of 64, indicating that most transactions are relatively low-to-mid value, while a smaller number of high-value purchases contribute significantly to total revenue. The Price by Category and Sales by Salesperson charts highlighted that phones, tablets, and watches generate the highest revenue, and that sales performance is uneven across sales representatives, with a small group consistently outperforming others. This insight is particularly important for identifying best practices and targeted training opportunities.

From a predictive analytics perspective, two classification models were evaluated to predict High vs. Low MRC customers. The Logistic Regression model achieved the strongest overall performance, with an accuracy of 97.66%, recall of 97.98%, and precision of 97.10%, demonstrating excellent balance in identifying high-value customers while minimizing misclassification. The Random Forest model, while still strong, achieved slightly lower accuracy at 96.68%, with recall of 95.95% and precision of 97.13%. These results suggest that Logistic Regression was more stable and better suited for this dataset, offering high interpretability alongside strong predictive power.

One of the most important insights from this project is the strong relationship between customer activation behavior, purchase patterns, and recurring revenue potential. Customers who both activated services and purchased devices or accessories showed higher MRC levels, making them ideal targets for cross-selling and upselling strategies. By combining interactive dashboards with machine learning predictions, the project delivers actionable intelligence that enables businesses to optimize sales strategies, improve employee performance management, and focus marketing efforts on customers with the highest long-term revenue potential.